Online Customer Purchase Intention

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Introduction To The Problem

Online shopping sites have millions of visitors that are served daily by the booming e-commerce. A primary issue with such sites, however, is the so-called window shopping, in which they have a large number of users browsing but only a small percentage of them make any purchase. The conditions leading to the conversion of a visitor into a buyer has come to be of great research interest to the business to enhance their marketing and relationship with their customers.

Developments in data analytics and machine learning have enabled the possibility to analyze user browsing patterns and predict patterns that reflect a buying intention. Predictive models may be used to determine the probability of a purchase within a user session based by scrutinizing variables including browsing period, page, type of device and nature of the user session. This allows companies to make decisions that are more informed in terms of targeted marketing, custom-created user experiences, and upgraded websites.

The given project is devoted to creating a machine learning model that will predict the presence of a purchase at the end of a browsing session as a visitor to a webpage. the analysis of behavioral and session-based characteristics is introduced to group sessions into purchase and non-purchase. The results would help explain consumer behavior and enable companies to enhance the rate of conversion and the entire online shopping experience.

Dataset Description

The data that has been used in this paper is the Online Shoppers Purchasing Intention Dataset which is obtained through the UCI Machine Learning Repository. It holds details on the browsing history of the user of an online shopping site, and is usually utilized in forecasting the likelihood of a customer to make a purchase on a given session. A dataset is made of each record, which is a user session, and the actions and features of a visitor visiting the site are recorded. The data set contains 17 various attributes that define user behavior and session attributes. These are variables that are connected with browsing (i.e. the amount of administrative, informational and product related pages accessed), time on various page categories, and patterns of interaction with the site. It also has categorical variables like browser type, operating system, region, month of visit and whether the visit was on a weekend or not.

The target variable in the dataset is the indication of a purchase at the end of the session and thus the problem is a binary classification one and the possible results are Purchase or No Purchase. With the help of such characteristics, machine learning algorithms can reveal similarities in user behavior and determine the probability of a visitor to make a purchase. This data is specifically valuable to analyzing customer behavior, enhancing target marketing strategy, and web design to maximize conversion rates.

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Administrative_Duration	Informational	Informational_Duration	ProductRelated	ProductRelated_Duration	BounceRates	ExitRates	PageValue	SpecialDay	Month	OperatingSystems	Browser	Region	TrafficType	VisitorType	Weekend
2	0	0	0	1	0	0.2	0.2	0	0	Feb	1	1	1	1	Returning_V	FALSE
3	0	0	0	2	64	0	0.1	0	0	Feb	2	2	1	2	Returning_V	FALSE
4	0	0	0	1	0	0.2	0.2	0	0	Feb	4	1	9	3	Returning_V	FALSE
5	0	0	0	2	2.666666667	0.05	0.14	0	0	Feb	3	2	2	4	Returning_V	FALSE
6	0	0	0	10	627.5	0.02	0.05	0	0	Feb	3	3	1	4	Returning_V	TRUE
7	0	0	0	19	154.2166667	0.01578947	0.024561	0	0	Feb	2	2	1	3	Returning_V	FALSE
8	0	0	0	1	0	0.2	0.2	0	0.4	Feb	2	4	3	3	Returning_V	FALSE
9	0	0	0	0	0	0.2	0.2	0	0	Feb	1	2	1	5	Returning_V	TRUE
10	0	0	0	2	37	0	0.1	0	0.8	Feb	2	2	2	3	Returning_V	FALSE
11	0	0	0	3	738	0	0.022222	0	0.4	Feb	2	4	1	2	Returning_V	FALSE
12	0	0	0	3	395	0	0.066667	0	0	Feb	1	1	3	3	Returning_V	FALSE
13	0	0	0	16	407.75	0.01875	0.025833	0	0.4	Feb	1	1	4	3	Returning_V	FALSE
14	0	0	0	7	280.5	0	0.028571	0	0	Feb	1	1	1	3	Returning_V	FALSE
15	0	0	0	6	98	0	0.066667	0	0	Feb	2	5	1	3	Returning_V	FALSE
16	0	0	0	2	68	0	0.1	0	0	Feb	3	2	3	3	Returning_V	FALSE
17	53	0	0	23	1668.285119	0.00833333	0.016313	0	0	Feb	1	1	9	3	Returning_V	FALSE
18	0	0	0	1	0	0.2	0.2	0	0	Feb	1	1	4	3	Returning_V	FALSE
19	0	0	0	13	334.9666667	0	0.007692	0	0	Feb	1	1	1	4	Returning_V	TRUE
20	0	0	0	2	32	0	0.1	0	0	Feb	2	2	1	3	Returning_V	FALSE
21	0	0	0	20	2981.166667	0	0.01	0	0	Feb	2	4	4	4	Returning_V	FALSE
22	0	0	0	8	136.1666667	0	0.008333	0	1	Feb	2	2	5	1	Returning_V	TRUE
23	0	0	0	2	0	0.2	0.2	0	0	Feb	3	3	1	3	Returning_V	FALSE
24	0	0	0	3	105	0	0.033333	0	0	Feb	3	2	1	5	Returning_V	FALSE
25	0	0	0	2	15	0	0.1	0	0.8	Feb	2	4	1	3	Returning_V	FALSE
26	0	0	0	1	0	0.2	0.2	0	0	Feb	2	2	4	1	Returning_V	TRUE
27	0	0	0	5	156	0	0.04	0	0	Feb	1	1	9	3	Returning_V	FALSE
28	64.6	0	0	32	1135.444444	0.00285714	0.009524	0	0	Feb	2	2	1	3	Returning_V	FALSE
29	0	0	0	4	76	0.05	0.1	0	0	Feb	1	1	1	3	Returning_V	FALSE

Methodology

The research presents a hybrid machine learning algorithm which is a combination of unsupervised and supervised machine learning to forecast whether an online browsing session leads to a purchase.

1. Data Preprocessing

The dataset has been cleaned and prepared by transforming the categorical variables (like browser, operating system and month) into numerical variables and converting the bool variables into the binary variables so that they can be handled by the machine learning algorithms.

2. K-Means Clustering Unsupervised Learning.

K-Means clustering was used and this categorized the browsing sessions as similar in terms of behavior. The number of optimum clusters was determined using the Elbow Method and Silhouette Score, which gave two clusters used in purchase and non-purchase sessions.

3. Pseudo Label Training and Supervision Model Training.

Pseudo labels were generated based on the cluster assignments as the output of the clustering was transformed into labeled data and then a Logistic Regression model was trained on these pseudo labels to classify browsing sessions as either purchases or non-purchases.

4. Model Evaluation and Importance Analysis.

The model was tested based on accuracy, precision, recall, and F1-score with high results with accuracy approximating 100 percent and few misclassifications in the confusion matrix. The significance of features was followed through the absolute coefficients of the logistic regression model. The factors that influenced were mainly the Month (particularly May), Traffic Type, Returning Visitor status and specific months like September and October as well as behavioral metrics like ProductRelated pages and Bounce Rate.

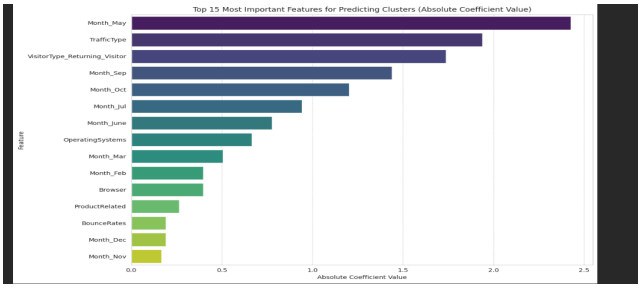
The results of model evaluation, and feature importance analysis are shown below.

```
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      3082
           1       0.99      0.99      0.99       617

    accuracy                           1.00      3699
   macro avg       0.99      1.00      0.99      3699
weighted avg       1.00      1.00      1.00      3699

Confusion Matrix:
[[3073    9]
 [   4  613]]
```



Societal Impacts

There can be some significant effects on society and the economy associated with the predictive models analyzing the behavior of online shoppers. The possibility of businesses using more effective and personalized marketing is one of the biggest advantages. Companies will be able to target promotions, discounts, and product recommendations more effectively by identifying users with a greater likelihood of making a purchase. This not only enhances the rate of conversion of businesses but also according to the consumer it gives them more relevant shopping experiences.

Website design and user experience may be also enhanced with the help of such models. With the knowledge about the browsing patterns that result in purchase, the companies will be able to optimize the navigation, product placement and the recommendation systems. The result is that it eases the online shopping experiences and causes less frustration to the users.

The other major impact is the contribution that it makes to the small and medium-sized businesses. Numerous online retailers, which are smaller, are largely dependent on online means to get their customers. The predictive analytics tools will assist these businesses to better use their marketing resources and compete with the bigger e-commerce sites by gaining a deeper understanding of customer behavior.

Nonetheless, ethical factors are also related to the usage of such models. User behavior data collection and analysis provokes the issue of privacy and data protection. Companies should also make sure that the information about their customers is managed in a responsible and regulation-friendly manner. Moreover, over-personalization or too much aggressive targeting might affect the consumer decision in a manner that might be perceived as intrusive unless done tactfully.

In general, used appropriately, purchase intention predictive models can make the business more efficient and customer experience more fulfilling, as well as contribute to the development of online commerce.

Conclusion

The project involved examining the hypothesis of predicting online shopping intent to purchase based on browsing session data by implementing a hybrid machine learning method. The combined methodology was an unsupervised learning with K-Means clustering, and a supervised learning with logistic regression. At the beginning, the Elbow Method and Silhouette Score were used by clustering methods to define the number of clusters and group sessions according to the similarity of behavior. Pseudo labels were then generated using these clusters and they were then used to train a supervised classification model.

The model trained had very good predictive performance with high accuracy, precision, recall, and F1-scores. The confusion matrix revealed that this model had minimal errors in classification and this data revealed that it is capable of quite effectively distinguishing between those sessions that have a tendency to lead to purchase and those that do not.

The analysis of feature importance also showed that the elements including the month of visit, type of traffic, returning visitor, and browsing behavior measurements are important factors that influence the purchase outcomes. Such insights can make businesses more aware of the customer behavior and maximize their marketing strategies and design of the website.

To sum up, the hybrid machine learning system employed in the current project offers an effective method of online purchase behavior prediction. Through a combination of the clustering/classification, the model can discover some unknown patterns in the browsing information and convert them into actionable information that can serve the business and consumer interests.